

QUAD PREAMP + 4-CHANNEL AMBISONIC RECORDER

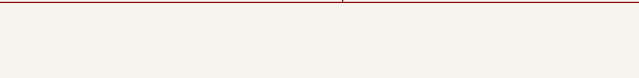
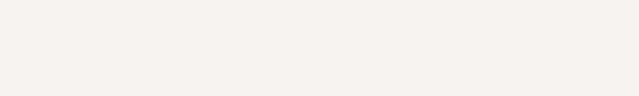
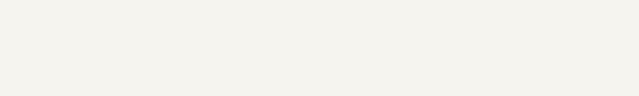
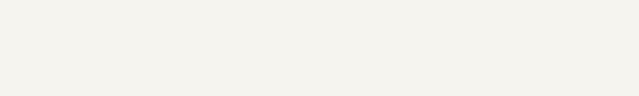
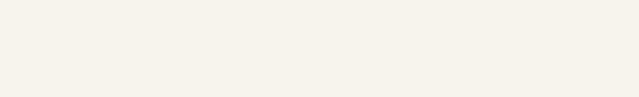
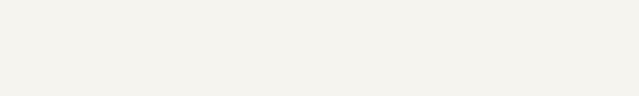
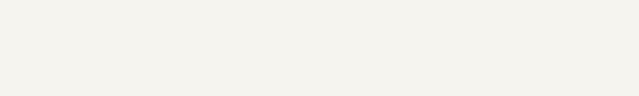
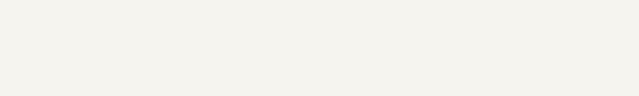
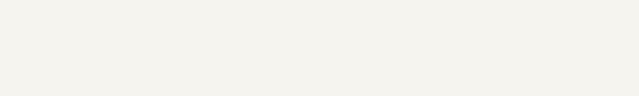
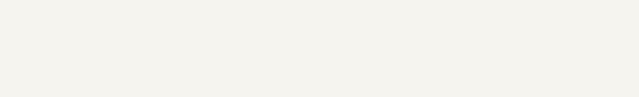
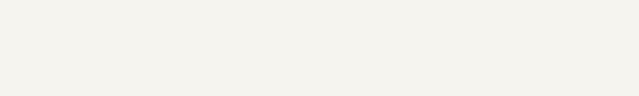
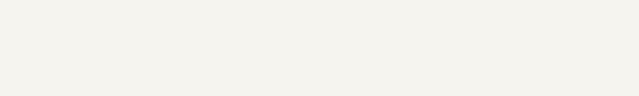
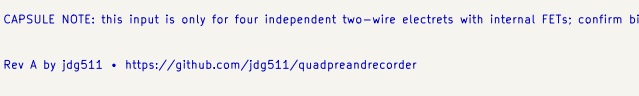
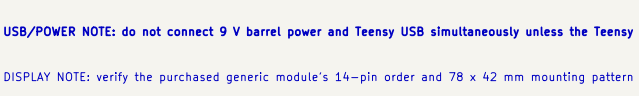
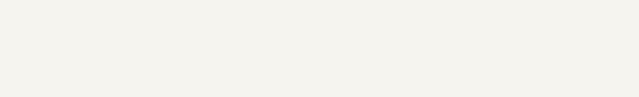
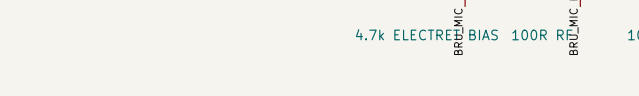
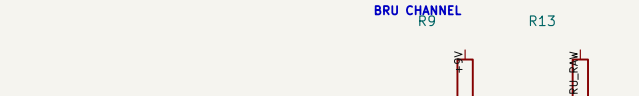
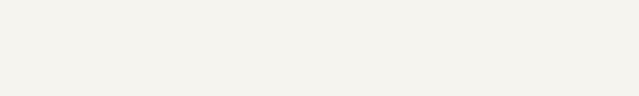
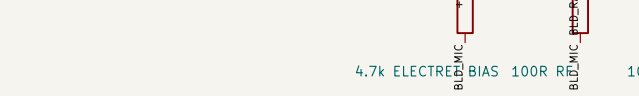
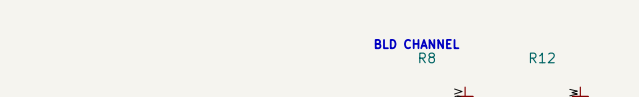
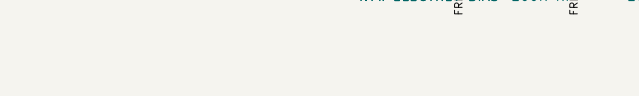
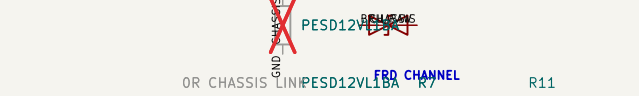
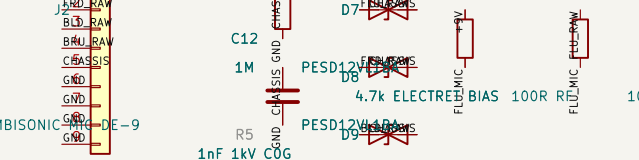
192 kHz / 24-bit capture • real-time binaural monitor • two-layer Rev A

POWER: protected 9 V input, 5 V buck, quiet 3.3 V audio rail, 4.5 V virtual ground

MICROPHONE INPUT + FOUR MATCHED CHANNELS

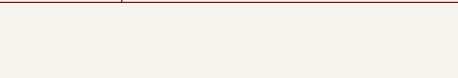
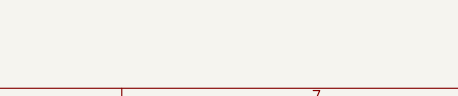
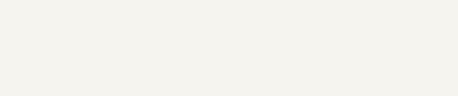
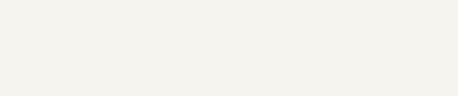
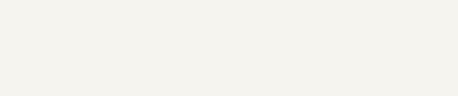
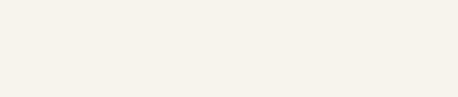
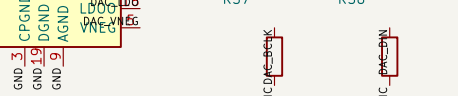
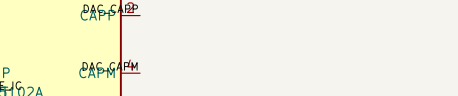
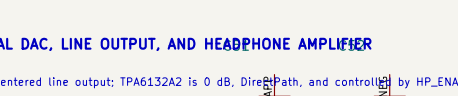
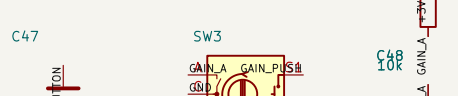
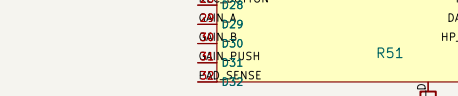
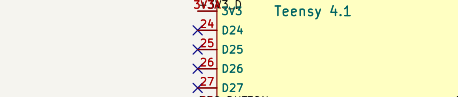
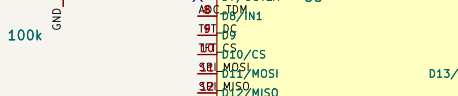
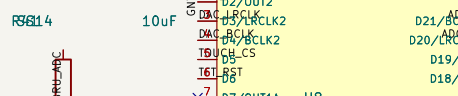
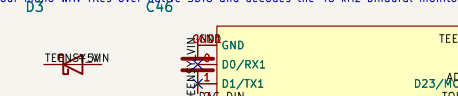
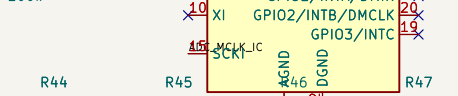
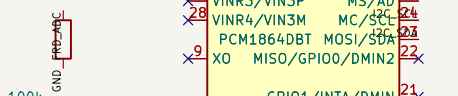
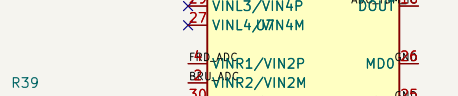
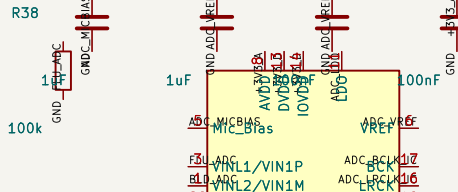
DE-9: 1 FLU+, 6 FLU return; 2 FRD+, 3 FRD return; 3 BLD, 6 BLD return; 4 BRU+, 9 BRU return; 5 shield

One PAD switch controls four CD4053 paths: HIGH-direct, LOW-FLU CHANNELS; 1454 fixed gain; 20.1 dB



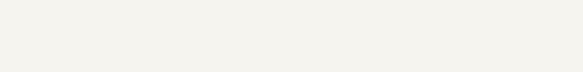
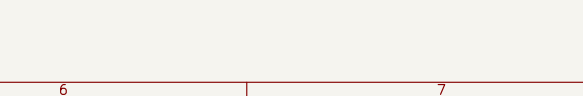
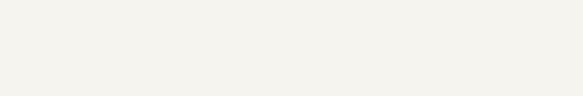
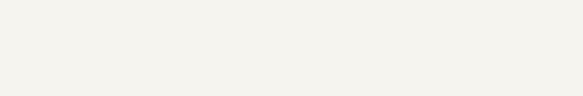
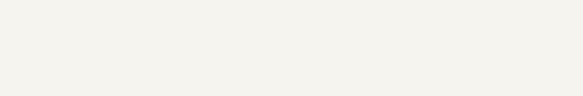
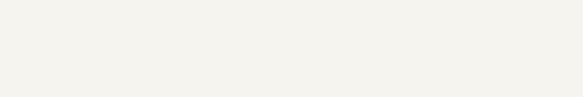
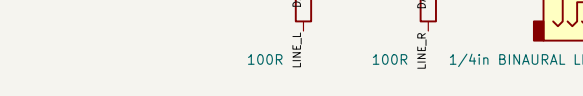
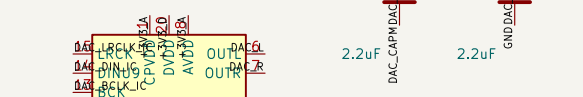
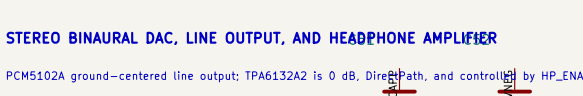
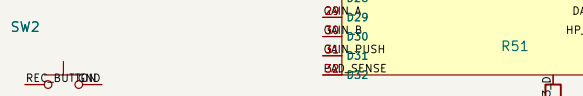
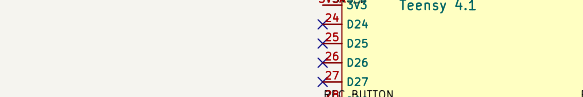
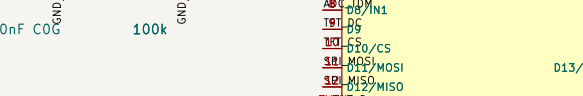
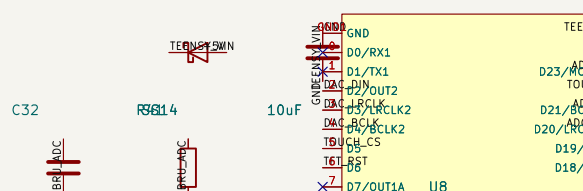
ADC + DIGITAL AUDIO

PCM1864: four single-ended inputs, common software PGA, 192 kHz 4-slot TDM



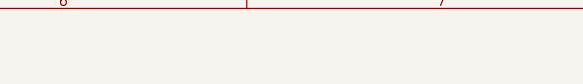
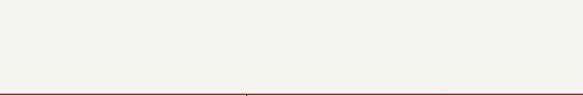
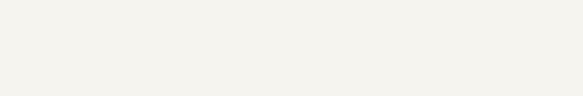
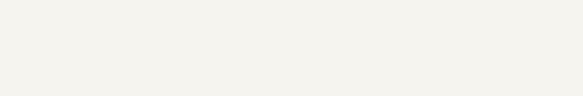
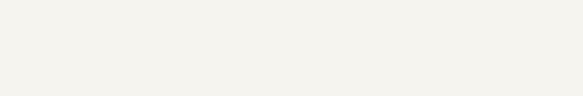
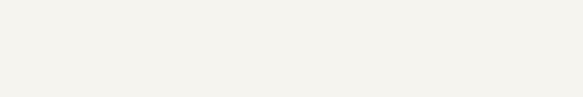
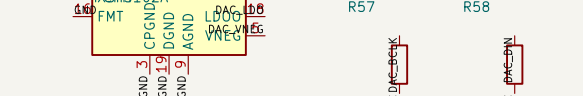
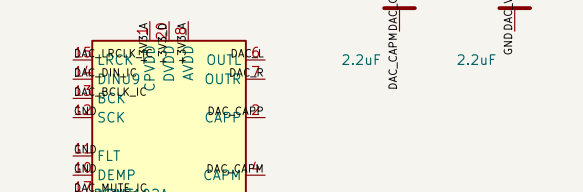
CONTROLLER, TOUCHSCREEN, RECORD CONTROLS

Teensy 4.1 records four 8000 WAV files over SPI SDIO and decodes the 48 kHz binaural monitor.



STEREO BINAURAL DAC, LINE OUTPUT, AND HEADPHONE AMPLIFIER

PCM5102A ground-centered line output; TPA6132A2 is 0 dB, Direct path, and controlled by HP_ENABLE.



USB/POWER NOTE: do not connect 9 V barrel power and Teensy USB simultaneously unless the Teensy VIN-VUSB link is cut.

DISPLAY NOTE: verify the purchased generic module's 14-pin order and 78 x 42 mm mounting pattern before enclosure drilling.

CAPSULE NOTE: this input is only for four independent two-wire electrets with internal FETs; confirm bias polarity and current.

Rev A by jdg511 • <https://github.com/jdg511/quadpreandrecorder>

Prototype before production; see hardware/requirements.md

Hammond 1590BB2 • two copper layers

Four 192 kHz / 24-bit simultaneous microphone channels

<https://github.com/jdg511/quadpreandrecorder>

jdg511

Sheet: /

File: QuadPreRecorder.kicad_sch

Title: Quad Preamp and Recorder — Four-Channel Ambisonic Front End

Size: A2 Date: 2026-07-19

Rev: A-prototype

KiCad E.D.A. 10.0.4

Id: 1/1